A Short Tutorial on Recurrence Relations

The concept: Recurrence relations are recursive definitions of mathematical functions or sequences.

For example, the recurrence relation

$$\begin{aligned} g(n) &= g(n-1) + 2n - 1 \\ g(0) &= 0 \end{aligned}$$

defines the function $f(n) = n^2$, and the recurrence relation

$$\begin{aligned} f(n) &= f(n-1) + f(n-2) \\ f(1) &= 1 \\ f(0) &= 1 \end{aligned}$$

defines the famous Fibanocci sequence 1,1,2,3,5,8,13,....

Solving a recurrence relation: Given a function defined by a recurrence relation, we want to find a "closed form" of the function. In other words, we would like to eliminate recursion from the function definition. There are several techniques for solving recurrence relations. The main techniques for us are the **iteration method (also called expansion, or unfolding methods or repeated substitution)** and the Master Theorem method.

Here is an example of solving the above recurrence relation for $g(n)$ using the iteration method:

$$\begin{aligned} g(n) &= g(n-1) + 2n - 1 \\ &= [g(n-2) + 2(n-1) - 1] + 2n - 1 \\ &= g(n-2) + 2(n-1) + 2n - 2 \\ &= [g(n-3) + 2(n-2) - 1] + 2(n-1) + 2n - 2 \\ &= g(n-3) + 2(n-2) + 2(n-1) + 2n - 3 \\ &= g(n-i) + 2(n-i+1) + ... + 2n - i \\ &= g(n-n) + 2(n-n+1) + ... + 2n - n \quad let i = n \\ &= 0 + 2 + 4 + ... + 2n - n \\ &= 2 + 4 + ... + 2n - n \\ &= 2 * n * (n+1)/2 - n \\ &= n^2 \end{aligned}$$

Applications: Recurrence relations are a fundamental mathematical tool since they can be used to represent mathematical functions/sequences that cannot be easily represented non-recursively. An example is the Fibanocci sequence. Here we are mainly interested in applications of recurrence relations in the design and analysis of algorithms.

Recurrence relations with more than one variable: In some applications we may consider recurrence relations with two or more variables. Here is an example recurrence relation with two variables.

$$\begin{aligned} T(m,n) &= 2*T(m/2,n/2)+m*n, \quad m>1, n>1 \\ T(m,n) &= n, \quad if m=1 \\ T(m,n) &= m, \quad if n=1 \end{aligned}$$

We can solve this recurrence using the iteration method as follows. Assume $m <= n$. Then

$$\begin{aligned} T(m,n) &= 2*T(m/2,n/2)+m*n \\ &= 2^2*T(m/2^2,n/2^2)+2*(m*n/4)+m*n \\ &= 2^2*T(m/2^2,n/2^2)+m*n/2+m*n \\ &= 2^3*T(m/2^3,n/2^3)+m*n/2^2+m*n/2+m*n \\ &= 2^i*T(m/2^i,n/2^i)+m*n/2^{(i-1)}+...+m*n/2^2+m*n/2+m*n \end{aligned}$$

Let $k = log_2 m$. Then we have

$$\begin{aligned} T(m,n) &= 2^k*T(m/2^k,n/2^k)+m*n/2^{(k-1)}+...+m*n/2^2+m*n/2+m*n \\ &= m*T(m/m,n/2^k)+m*n/2^{(k-1)}+...+m*n/2^2+m*n/2+m*n \\ &= m*T(1,n/2^k)+m*n/2^{(k-1)}+...+m*n/2^2+m*n/2+m*n \\ &= m*n/2^k+m*n/2^{(k-1)}+...+m*n/2^2+m*n/2+m*n \\ &= m*n*(2-1/2^k) \end{aligned}$$

Given a recurrence relation:

$$\begin{aligned} T(n) &= T(n-1)+O(n) \\ T(1) &= 1 \end{aligned}$$

(You may also write simply $T(n) = T(n-1)+n$ if you think of $T(n)$ as the the number of multiplications)

By a straightforward expansion method, we can solve T(n) as:

$$\begin{aligned} T(n) &= T(n-1) + O(n) \\ &= (T(n-2) + O(n-1)) + O(n) \\ &= T(n-2) + O(n-1) + O(n) \\ &= T(n-3) + O(n-2) + O(n-1) + O(n) \\ &= T(1) + O(2) + ... + O(n-1) + O(n) \\ &= O(1 + 2 + ... + n - 1 + n) \\ &= O(n^2) \end{aligned}$$

Another recurrence relation:

$$\begin{aligned} T(n) &= T(n/2) + O(n/2) \\ T(1) &= 1 \end{aligned}$$

We can solve it as:

$$\begin{aligned} T(n) &= T(n/2) + O(n/2) \\ &= (T(n/2^2) + O(n/2^2)) + O(n/2) \\ &= T(n/2^2) + O(n/2^2) + O(n/2) \\ &= T(n/2^3) + O(n/2^3) + O(n/2^2) + O(n/2) \\ &= T(n/2^i) + O(n/2^i) + ... + O(n/2^2) + O(n/2) \\ &= T(n/2^{logn}) + O(n/2^{logn}) + ... + O(n/2^2) + O(n/2) * \\ &= T(1) + O(n/2^{logn}) + ... + O(n/2^2) + O(n/2) \\ &= 1 + O(n/2^{logn} + ... + n/2^2 + n/2) \\ &= 1 + O(n * (1/2^{logn} + ... + 1/2^2 + 1/2) \\ &= O(n) * * \end{aligned}$$

* - We stop the expansion at $i = logn$ because $2^{logn} = n$
** - $1/2^{logn} + ... + 1/2^2 + 1/2 \leq 1$